Reg. No. | | | | | | | | | | |

[18CS5SD01/18MA5DE11/18CT5DE11/18AI5DE11]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021
THEORY OF COMPUTATION

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Assume you are going to organize a game where you don't want two partners to sit together of same gender, that is, every boy should sit next to a girl. Formulate the game as a formal language and design a deterministic finite automat for the same. **10**

b. Prove by giving an example that, if the machine M is an Non deterministic Finite Automata that recognizes language L, swapping the accept and non-accept states in M doesn't necessarily yield a new NFA that recognizes $\bar{L}$. **04**

2. a. **i.** Construct an equivalent FA for the given regular expression (0+1)*(00) **06**

ii. Prove or disprove the following statement: If L1 and L2 are not regular languages, then L1 ∪ L2 is not regular. **04**

b. For the given DFA, construct equivalent regular expression using state elimination method. A start and C final state **04**

	a	b
A	B	C
B	-	A
C	A	B

3. a. How do you define ambiguity in the grammar? Is the grammar **09**

S -> SAS | 0

A -> ASA | 1

(S is the start symbol) is ambiguous. Justify your answer with an example.

b. What is Context Free Grammar? Give a CFG for the language { $a^i b^j c^k$ | i, j, k ≥ 0 and i + j = k } **05**

OR

4. a. Show that the following language is not context free using pumping lemma L={ $a^m b^m c^m$ | m >=1} **07**

b. How do you identify that the grammar is in Chomsky Normal form? Is the following CFG is CNF? If not rewrite the grammar into an equivalent CFG in Chomsky normal form **07**

S → BSB | B | ε

B → 00 | ε

5.	**a.**	**i.**	Describe how to find equivalent PDA for the given CFG? And Apply the same for the given CFG and design equivalent PDA.S → XS | ε , A → aXb | Ab | ab	**08**
		ii.	Differentiate deterministic and non-deterministic push down automata.	**03**
	b.		A PDA is more powerful than a finite automaton. Justify this statement	**03**
			OR	
6.	**a.**		Design a Turing machine which accepts a set of strings w made up of a's and b's and WW^R where W^R is the reverse string of W. Determine whether it is deterministic or non- deterministic in nature?	**10**
	b.		Describe the architecture of a 3- tape Turing machine and explain the functionality of the same.	**04**
7.	**a.**		Describe briefly recursive and recursive enumerable languages.	**08**
	b.		Describe the classification of grammars with the help of Chomsky hierarchy.	**06**
			OR	
8.	**a.**	**i.**	Does the following instance of PCP have a solution: $A = (ab, b, aba, aa)$ and $B = (abab, a, b, a)$?	**05**
		ii.	Describe Turing Machine Halting Problem?	**04**
	b.		What is Linear bounded automata? List and explain 3 unique applications of LBA.	**05**